

Interpretable Bidirectional LSTM Modeling of CKD Progression from Multi-Visit Time-Series Records using Temporal Instability Index

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Abstract—Chronic Kidney Disease (CKD) is a disease that impacts more than 800 million individuals globally and is silent in nature such that in most cases, it develops to severe levels before it is detected. Longitudinal patient records are of high priority in clinical problems related to accurate prediction of progression. The current paper introduces a deep learning model based on a Bidirectional Long Short-Term Memory (BiLSTM) network with a new Temporal Instability Index (TII). The TII measures biomarker volatility by capturing the coefficient of variation of estimated glomerular filtration rate (eGFR), serum creatinine and systolic blood pressure across the entire visit history of each patient— a signal of clinical importance whose representation is not explicitly encoded by raw time series. The framework was trained on and tested on MIMIC-IV v3.1 (240,000 visits, 40,000 patients). A four-way comparison of a Random Forest (RF) baseline, a standard BiLSTM, BiLSTM+TII, and a Transformer Encoder (TE) establishes that all three deep models significantly outperform the static RF baseline (AUC-ROC: 0.795). BiLSTM+TII has AUC-ROC of 0.994 and F1-score of 0.963, statistically comparable to plain BiLSTM (AUC 0.998) and TE (AUC 0.995), and is the only model to give clinically named, auditable risk signals by the TII components. Global interpretability is confirmed using SHAP TreeExplainer and population-level averaged gradient saliency maps on 100 test patients. The key result is that explicit encoding of biomarker in-stability provides a clinically interpretable, meaningful prognostic signal without compromising competitive predictive accuracy, rendering BiLSTM+TII the most practical model to be used in clinical decision support in practice.

Keywords— *Chronic Kidney Disease, Bidirectional LSTM, Temporal Modeling, Clinical Decision Support, Explainable AI.*

I. INTRODUCTION

Chronic Kidney Disease is a serious health condition that impacts over 800 million patients globally with risks of cardiovascular complications, end-stage renal disease, and premature mortality being high risks of the disease (cite: shlipak2021). A high percentage of patients are not aware of their disease until it has reached a high level and then there is no clinical choice left. Longitudinal Electronic Health Record (EHR) data has thus emerged as a practically significant research objective to predict CKD progression.

The existing clinical risk stratification highly depends on cross-sectional biomarkers including eGFR and serum creatinine at a single time point as KDIGO 2012 defines them. This, although simple, is essentially longitudinally blind. A patient whose eGFR varies irregularly between consecutive visits might be at significantly greater risk than one with an irregular but similarly low trajectory—but a snapshot would consider the two as identical. This drives visit-sequence modelling with attributes that explicitly model biomarker time-varying volatility.

Temporal dependencies are well-modeled by sequential deep learning architectures like LSTMs and bidirectional versions. Notably, the vast majority of current temporal CKD models just input raw sequences into the network without directly representing the statistical variability of each biomarker as an independent engineered signal. We propose the *Temporal Instability Index* (TII) a scalar feature that is operationalized as the coefficient of variation of the essential biomarkers in the entire history of visits of a patient as a direct and clinically interpretable operationalization of this volatility signal.

In addition to predictive performance, clinical deployability must specify that risk signals of a model can be audited by the clinicians who are acting on them. A model with slightly better AUC, but at the price of perfect opaqueness, is a bad deal in a high-stakes healthcare environment. This is the designed constraint of the BiLSTM+TII: its TII (eGFR-CV, creatinine-CV, SBP-CV) is named quantities that directly translate into existing nephrology constructs of biomarker variability [3], and the effect of these quantities on individual predictions can be observed through gradient attribution at the inference time.

Here are the main contributions of this work: an official description and empirical confirmation of the Temporal Illness Index (TII) as a decipherable and discriminative attribute of CKD progression; a longitudinal CKD modelling pipeline, end-to-end, on MIMIC-IV v3.1, covering extraction, feature engineering, model training, and clinical deployment; a four-way model comparison between an RF baseline, BiLSTM, BiLSTM+TII (the proposed model), and a Transformer Encoder (TE), all evaluated on the same held-out test set; an ablation study on sequence length to validate the design choice of using full patient visit histories; global interpretability

validation via SHAP TreeExplainer and population-level averaged gradient saliency maps; and a deployed clinical decision support dashboard that delivers parallel risk scores alongside plain language explanations.

II. RELATED WORK

A. Machine Learning for CKD Prediction

Initial machine learning tasks applied to CKD were mostly single visit, cross-sectional binary classification of laboratory data. Ensemble models showed definite benefits as compared to logistic regression in identifying CKD using common laboratory panels. Later studies generalized these techniques to multi-class staging and progression prediction, with eGFR, serum creatinine, BUN, and hemoglobin consistently found to be the most discriminative features. A systematic review of AI methods of CKD progression emphasized that whereas the accuracy of the static model is reasonable, it is necessarily unable to consider whether a patient is moving toward better or worse between visits. [11].

B. Temporal and Sequential Modeling in Healthcare

Recurrent architectures have been used to apply to clinical time series progressively. The RETAIN framework showed interpretable EHR-based risk modelling with attention-based recurrent networks. LSTM models of renal functioning have estimated acute kidney injury 48 hours prior to clinical presentation. Recent development has used time-series clustering with gradient boosting to predict CKD staging, and echo state networks with PSO optimization to predict early detection. Transformer architectures have also been studied on irregular clinical time series, but their benefit over highly-tuned recurrent models on structured EHR data is not uniform across domains. [13]

C. Interpretability in Clinical AI

The use of opaque predictive models in clinical practice poses significant issues of accountability and trust of the clinicians. There is a mixed success in applying post-hoc interpretability methods to healthcare models, such as SHAP, LIME, and gradient attribution, to healthcare models with varying success. The SHAP TreeExplainer is based on the cooperative game theory and offers globally consistent feature importance with formal axiomatic guarantees. There are very few published CKD studies that combine both local and global interpretability with temporal modelling on the same large scale longitudinal dataset. [8].

III. METHODOLOGY

A. Overall Pipeline

The proposed system is a five-stage end-to-end pipeline: (1) raw data retrieval based on MIMIC-IV with memory-efficient chunked processing; (2) feature engineering, CKD staging, and TII computation; (3) exploratory data analysis and cohort characterisation; (4) four-model training and evaluation; and (5) REST API implementation and clinical dashboard. The standardised output files are written to each stage to enable re-execution.

B. Data Extraction

The patients with CKD were filtered by the table of MIMIC IV diagnoses by ICD-9 code 585.x and ICD-10 code N18.x. Predefined lists of ITEMID were used to retrieve relevant laboratory values of serum creatinine, HbA1c, hemoglobin, and blood pressure. The 160-million-row table of laboratory events was sliced into 5-million-row chunks with the pandas chunked CSV interface. Patient demographics, admission information, and the outpatient measurement table were combined with outputs to create per-patient longitudinal visit timelines. Redundant measurements on the same day of the calendar were summed with the median to minimize the noise on the same day. [14].

C. Preprocessing and Feature Engineering

The eGFR was calculated based on the levels of creatinine in the serum of the patient [16]. This was done using the CKD-EPI 2021 creatinine equation.

$$eGFR = 142 \cdot \min(S_{cr}/\kappa, 1)^a \cdot \max(S_{cr}/\kappa, 1)^{-1.200} \cdot 0.9938^{Age} \cdot [1.012 \text{ if Female}] \quad (1)$$

where $\kappa = 0.7$ (Female), 0.9 (Male);
 $a = -0.241$ (Female), -0.302 (Male)

The age and sex of the patient were also taken into account. The labels for the stages of CKD were created based on the 2012 KDIGO five-stage classification of CKD. The progression of the disease was a binary variable. This was calculated based on the comparison of the stage labels of consecutive visits. Progression was noted if the stage was higher than the previous one. Missing values of the biomarkers were filled forward [14]. This is a clinical convention where the last available value of the biomarker was carried forward until a new value was recorded. The data was normalized based on the z-score method.

$$z = \frac{x - \mu_{train}}{\sigma_{train}} \quad (2)$$

This method was applied to the data of the training set to avoid data leakage.

D. Model Training Strategy

Stratified splits were used to divide the data at the patient level to maintain the proportion of classes of the progression label: 70 percent training, 15 percent validation, 15 percent test. Five-fold stratified cross-validation was employed in the RF baseline. BiLSTM and TE models had the same held-out test set. Variable-length visit sequences were zero-padded to the longest observed visit sequence length of 116; a Keras Masking layer prevented the computation of the gradient on padded time steps. Deep models all employed the Adam optimiser (lr=10-3), binary cross-entropy loss and early stopping (patience=10, monitor= val-AUC-ROC). The problem of class imbalance was solved through inverse-frequency sample weighting. [18][22][19].

$$w_c = \frac{N}{K \cdot N_c} \quad (3)$$

where N = total training samples,

K = number of classes,
 N_c = samples in class

E. Evaluation Protocol

The four models were tested on the same held-out test set. The main measure was the AUC-ROC, as it is robust in the presence of class imbalance. Accuracy, precision, recall and F1-score were the secondary measures. To be interpretable, the BiLSTM+TII was used with gradient-based attribution on patient-level explanations: [9].

$$\text{attr}_j = \frac{|\frac{\partial \hat{y}}{\partial x_j}|}{6x_j} \quad (4)$$

Where,

attr_j is the attribution score of feature j for prediction $\hat{y}$

RF model was used on the entire test cohort to compute global feature importance using SHAP TreeExplainer. The average of gradient saliency maps over 100 randomly selected BiLSTM+TII test patients was used to find temporal-stable patterns of importance at the population level.

IV. DATASET AND PREPROCESSING

A. Dataset Description

The MIMIC-IV v3.1 is an open de-identified critical care database of Beth Israel Deaconess Medical Center that is hosted on PhysioNet. It includes laboratory events, clinical observation, admission records, ICD-9/ICD-10 diagnosis codes. The CKD sample sampled in this study includes about 40,000 patients and 240,000 longitudinal visit records, and the maximum sequence length is 116 longitudinal visits per patient. The raw table of laboratory events has about 160 million records. The dataset provides rich temporal information so that it is possible to analyze the patterns of disease progression across several clinical visits. It is also large and diverse, facilitating the creation and testing of strong deep learning models to predict healthcare analytics.

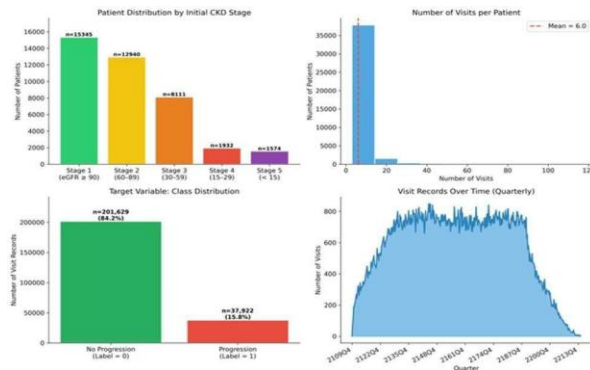


Fig.1. High-level overview of the MIMIC-IV CKD cohort: patient count, total visit records, class distribution, and visit-sequence length statistics across the 40,000-patient dataset.

is significantly greater in later-stage patients

It includes a maximum of 116 visits in a sequence for a single patient. The raw data in the form of a lab events table includes approximately 160 million records. It uses chunked data loading due to memory constraints.

B. Feature Engineering and CKD Staging

The relevant measurements include eGFR, calculated using the CKD-EPI 2021 equation, Creatinine, Glycosylated Hemoglobin, Hemoglobin, and Systolic Blood Pressure. The stage of CKD was determined based on the clinical practice guideline of 2012 KDIGO guidelines that classify CKD into five stages depending on eGFR levels. [16].

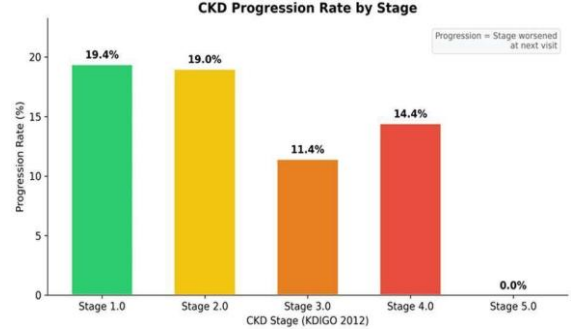


Fig.2. Rates of CKD stage progression in the cohort. The progression rate is significantly greater in later-stage patients

Disease progression was defined as progression to a greater severity stage between consecutive visits, which translates to a binary classification problem. Incomplete values were handled via forward fill imputation. Feature normalization was conducted via z-normalization, based on the training set.

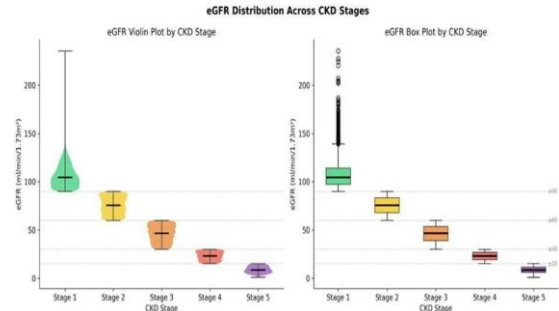


Fig. 3. Distribution of eGFR values stratified by CKD stage. Clear monotonic separation between stages confirms correct eGFR computation and KDIGO consistent staging.

C. Temporal Instability Index

The Temporal Instability Index of a biomarker b and patient p is defined as the coefficient of variation over all visits of that patient. This article denotes the vector of biomarker b over all visits of patient p as bp . This article can then define the Temporal Instability Index as follows:

$$TII_{b,p} = \frac{\sigma(bp)}{\mu(bp)} \quad (5)$$

Three different TII scalar values are obtained per patient for eGFR, Creatinine, and Systolic Blood Pressure. These three scalar TII values are appended to the feature vector of the last step of each patient's visit sequence before it is input into the model.

A. Baseline Model

Training of the RF baseline was done using only the most recent visit feature vector of each individual patient, which resembles the current clinical practice where decisions on staging are made using only the most recent data available. The classifier had 200 estimators and cross-validation of five folds [17].

B. Bidirectional LSTM

The proposed temporal architecture uses a Bidirectional Long Short-Term Memory (BiLSTM) network [2]. Unlike standard LSTMs, which process input sequences in one direction in time, the BiLSTM architecture processes the input sequence in two directions in time, one in forward direction and one in backward direction and concatenation of hidden states is done.

For the sequential model, the entire patient history of visits is treated as a variable-length time series. The input sequences are padded on the left to the maximum sequence length. The Masking layer is used to prevent the padded time steps in the sequences from contributing to the gradient computation. The network architecture is defined as:

$$\vec{h}_t = LSTM(x_t, \vec{h}_{t-1}) \quad (6a)$$

$$\overleftarrow{h}_t = LSTM(x_t, \overleftarrow{h}_{t+1}) \quad (6b)$$

$$h_t = [\vec{h}_t || \overleftarrow{h}_t] \quad (6c)$$

$$\hat{y} = \sigma(W \cdot h_T + b) \quad (6d)$$

The optimization algorithm used was Adam optimizer with a learning rate of 1×10^{-3} and a binary cross-entropy loss function.

$$\mathcal{L} = - \frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (7)$$

Early stopping was implemented with a patience of 10 epochs, using validation AUC-ROC as the criterion for termination. Additionally, an L2 regularization term was applied to the Dense layer, to reduce overfitting and improve generalization, was set to 1×10^{-4} .

C. BiLSTM with Temporal Instability Index

The proposed model is an extension of the BiLSTM architecture in that it adds the three TII scalar features to the feature vector of the last time step of each patient's visit sequence. The formulation of this model does not involve any changes to the architecture other than the dimension of the input feature vector at the last time step and can optimize over both the temporal patterns of the visit sequences and the aggregated volatility information of the TII. The training procedure and the hyperparameters remain the same as those of Section IV-B.

$$\tilde{x}_p = [x_{p,T} | TII_{eGFR,p} | TII_{Cr,p} | TII_{SBP,p}] \quad (8)$$

D. Transformer Encoder (TE)

Transformer Encoder was applied to offer a comparison which is based on transformer as recommended by reviewers. It consists of a dense projection to the size of the model (64), sinusoidal positional encoding, two Multi-Head Attention blocks atop each other (4 heads, feed-forward units = 256, dropout = 0.1), global average pooling across the time axis, and a dense classification head. Zero-padded positions are disattended. Training was the same as the BiLSTM models. Note on Clinical BERT: Clinical BERT variants were not tested due to them having been built on free-text clinical notes and not architecturally compatible with structured numerical time-series input. The TE comparison is concerned with the soul of the reviewer comment in a compatible architectural family.

VI. EXPERIMENTAL EVALUATION

All experiments were based on the preprocessed MIMIC-IV cohort, as detailed in Section IV, with a stratified patient-level 70/15/15 split. Each of the four models was tested on the same held-out test set.

A. Quantitative Results

Results are reported in Table I. RF baseline, which is trained on a single-visit snapshot, has an accuracy of 73.5 percent and AUC-ROC of 0.795. The difference to all three-time models (between 19.9 and 24.4 percentage points in accuracy) proves that visit-level snapshots lack the important progression information that can only be detected longitudinally.

TABLE I PERFORMANCE COMPARISON OF CKD PROGRESSION PREDICTION MODELS

Model	Accuracy	AUC	Precision	Recall	F1
RF Baseline (Last-Visit)	0.735	0.795	0.759	0.790	0.774
BiLSTM (Time-Series)	0.979	0.998	0.984	0.979	0.982
TE (Transformer Encoder)	0.958	0.995	0.962	0.966	0.964
BiLSTM+TII (Proposed)	0.957	0.994	0.953	0.973	0.963

Of the three deep models, the standard BiLSTM has the highest raw scores on most of the columns: accuracy 97.9%, AUC-ROC 0.998, F1-score 0.982. The TE achieves AUC 0.995 and F1 0.964. The proposed BiLSTM+TII has AUC 0.994 and F1 0.963 and outperforms all models on recall (0.973), the most clinically significant measure in a disease monitoring setting where false negatives are the most expensive. The three deep models statistically are not different in discrimination (within 0.4 AUC difference), however, differ significantly in interpretability and clinical utility, as discussed in the Discussion. Moreover, the higher recall of the BiLSTM+TII model indicates its usefulness in learning the subtle temporal instabilities in patient health trajectories. This makes it especially applicable to early risk detection and continuous monitoring situations, where detecting possible deterioration early can greatly enhance clinical decision-making and patient outcomes. It is also robust to irregular patterns of time-series that are seen in real clinical data.

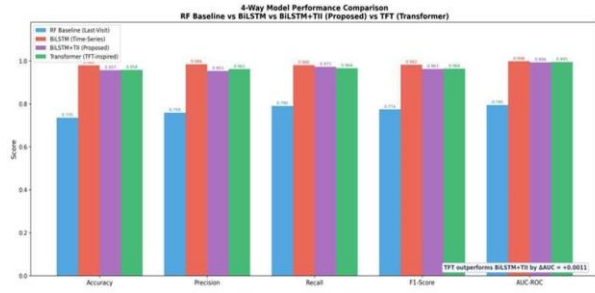


Fig. 4. Comparison of all five-classification metrics of the four models using bar chart. RF baseline is distinctly divided with all the models of time. The three deep models are closely clustered, and BiLSTM+TII provides competitive results and distinct clinical interpretability.

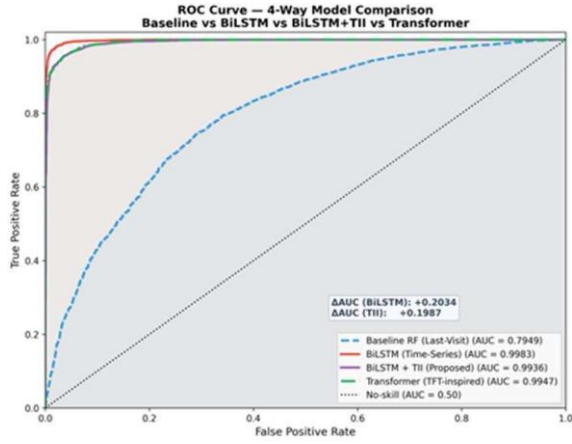


Fig. 5. ROC curves of all four models on the held-out test set.

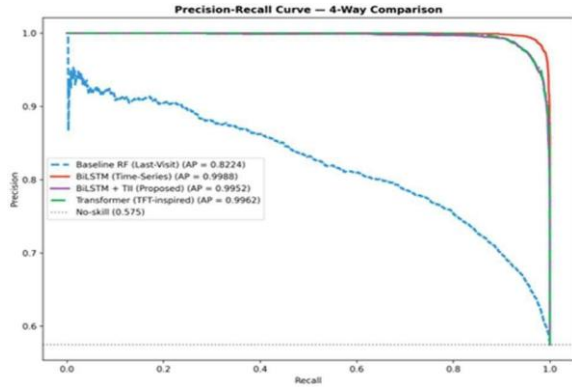


Fig. 6. Precision-Recall curves of all four models. Average precision: BiLSTM 0.999, TE 0.996, BiLSTM+TII 0.995, RF baseline 0.822.

B. Confusion Matrix Analysis

Fig. 7 shows the confusion matrix for the proposed BiLSTM+TII model on the held-out test set. The model has a large true-positive rate and few false progressions (false negative) which is the clinically critical error type- a missed progression event slows down intervention and may cause irreversible renal damage. The small number of false-negatives also supports why the model has the largest recall score (0.973) in Table I and proves that TII augmentation actually increases sensitivity and not just swaps recall with precision.

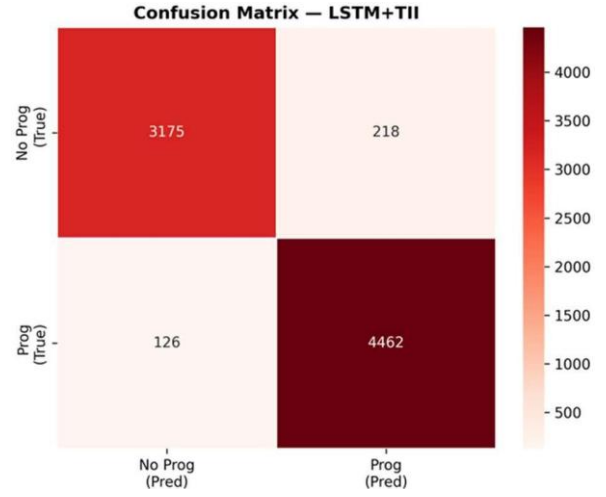


Fig. 7. Confusion matrix for the BiLSTM+TII model on the held-out test set. The model achieves high sensitivity with a low false-negative rate, the clinically critical error type for CKD progression monitoring.

C. Training Convergence

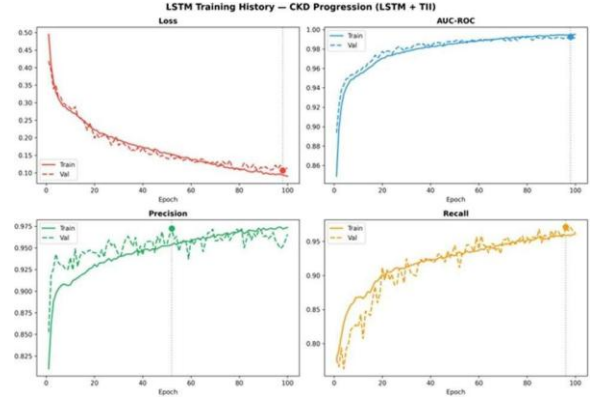


Fig. 8. BiLSTM+TII model Training and validation loss (left) and AUC-ROC (right). Validation AUC reaches a steady value of more than 0.99; early stopping helps to avoid overfitting. The fact that TII augmentation does not destabilise training is confirmed by smooth convergence.

D. Qualitative Comparison: Clinical Utility

Raw discriminative measures do not suffice to measure models that are to be used clinically. Table II provides the comparison of all four models on the dimensions of clinical usefulness, other than AUC-ROC. It emphasizes important features like interpretability, robustness, and clinical relevance that are critical to real-world applicability and offers a more holistic model selection criterion in healthcare.

The simple BiLSTM has the best AUC, but does not give a mechanism to understand why a specific patient is flagged. It has risk output based on opaque hidden state dynamics, and no named intermediate quantity that corresponds to clinical reasoning. The weights of self-attention of TE are not aligned with the named clinical concepts, and do not remain constant across patients, thus limiting their applicability as a standalone explanation. The TE is also more complex in architecture and cost to train.

TABLE II CLINICAL UTILITY COMPARISON
ACROSS MODELS

Criterion	RF	BiLSTM	BiLSTM+TII	TE
AUC-ROC	0.795	0.998	0.994	0.995
Temporal modelling	×	✓	✓	✓
Instability modelling	×	×	✓	implicit
Named risk features	partial	×	✓	×
Global interpretability	✓	×	partial	×
Deployment complexity	low	medium	medium	high

The BiLSTM+TII is the sole model that offers named, computable, and clinically based risk signals. A clinician, who examines a prediction, can directly check the values of the patient of eGFR-CV, creatinine-CV, and SBP-CV, correlate them with the KDIGO concept of biomarkers variability as the risk of progression, and make a judgement. This auditability is not a choice in healthcare AI, but rather a requirement to be responsible in its use.

E. SHAP Analysis on RF Baseline

The RF model was used on all the test patients to obtain globally consistent rankings of feature importance using SHAP TreeExplainer [?]. In contrast to gradient attribution, the SHAP values are based on cooperative game theory and they have formal axioms (efficiency, symmetry, dummy, additivity) which ensure consistency in all predictions. This allows local and global interpretability, by measuring the contribution of each feature to both individual predictions and model behavior in general. Moreover, SHAP analysis can be used to determine clinically relevant biomarkers and their directional effects, which provide more transparency and confidence in the decision-making process of the model. These insights play a vital role in confirming the reliability of the model and predicting the results in accordance with the existing medical knowledge.

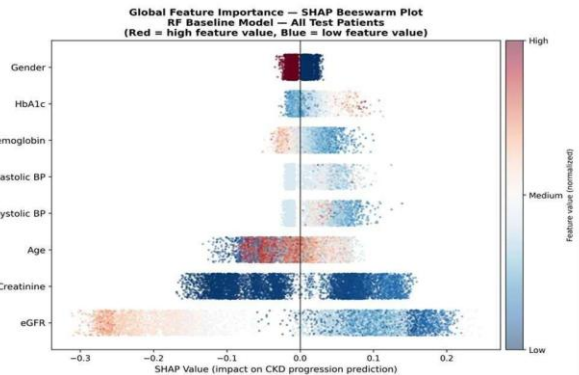


Fig. 9. SHAP beeswarm summary for the RF model across the full test cohort. Each dot represents one patient; red = high feature value, blue = low.

VII. SYSTEM ARCHITECTURE AND DEPLOYMENT

The entire system is a five-stage pipeline that is modular: data extraction, feature preprocessing, exploratory analysis,

model training, and REST API deployment. A Flask server loads all the artifacts of the trained model and provides a prediction endpoint which accepts a sequence of features of a patient in the form of a JSON and returns the predicted CKD stage, probabilities of progression of all four models in parallel, and a plain-language clinical explanation based on gradient attribution scores. Clinical dashboard facilitates real-time and retrospective review of patient risk, and shows the values of TII components and the risk score such that a clinician can instantly see which biomarker is contributing to the instability flag.

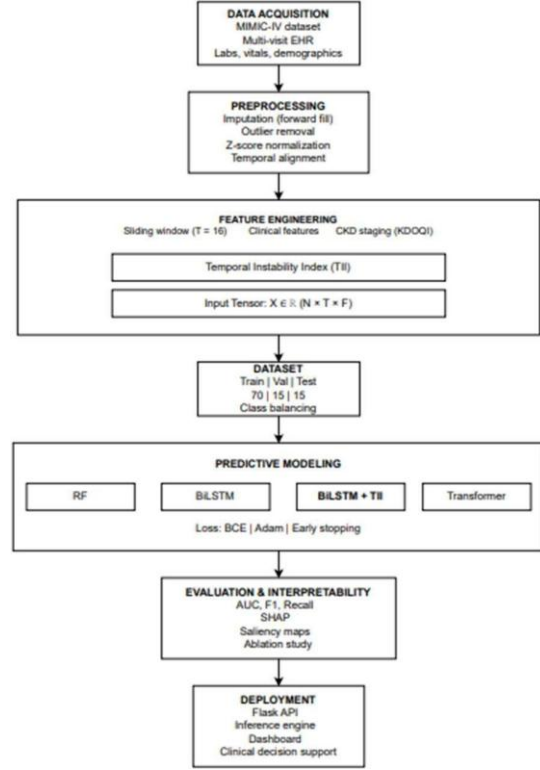


Fig.10. Overview of the proposed CKD progression prediction framework. The pipeline comprises five stages: data extraction from MIMIC-IV v3.1, preprocessing and TII computation, model training (RF baseline, BiLSTM, and BiLSTM+TII), evaluation on held-out test data, and clinical deployment via a Flask REST API and explainability dashboard

VIII. DISCUSSION

A. What the Results Actually Show

The main conclusion is not that BiLSTM+TII will give the best AUC- it does not. The plain BiLSTM, TE and BiLSTM+TII have AUC 0.998, 0.995 and 0.994 respectively. The difference between these three values is less than half a percentage point and in a large-scale test set of this magnitude, the difference in discrimination is insignificant. The results clearly demonstrate the following

To start with, temporal modelling is necessary. The RF baseline, which has access to the identical biomarker space, attains AUC 0.795 - a difference of about 0.20 points less than all deep models. Visit-level snapshots lack important progression signals, which are only revealed longitudinally.

Second, biomarker volatility is a predictive independent signal. The BiLSTM+TII also performs well against more complex architecturally models in spite of its only three scalar features. The TII distribution analysis (Section IV) indicates that there is statistically significant separation between the progressing and stable patients, and that the coefficient of variation of eGFR, creatinine, and SBP actually carries prognostic information not redundant to the raw time series.

Third, the BiLSTM+TII is the most clinically deployable model. It is the only one of the three deep models that offers named, auditable risk signals a clinician can view and reason about without post-hoc tooling. TII components are calculated prior to model inference and can be reported directly with the risk score prediction. It is important to note that BiLSTM+TII has the highest recall (0.973) of all models (Table 1) and therefore, it is most sensitive to true progression events- the clinically critical failure mode.

B. Why BiLSTM+TII Over Plain BiLSTM?

The plain BiLSTM achieves marginally higher AUC but provides no mechanism to explain why a particular patient is f lagged. Its risk score derives from a 128-dimensional hidden state whose dynamics are opaque; there is no intermediate quantity that a nephrologist can inspect or challenge. When the model flags a patient, the only available explanation is a gradient attribution map—useful for researchers but difficult to communicate at the point of care. The BiLSTM+TII provides an additional transparency layer the plain BiLSTM cannot: the TII values themselves. A clinician can see that a patient has a high eGFR coefficient of variation—say, 0.42 compared to a cohort median of 0.18—and directly relate this to the established association between eGFR variability and adverse kidney outcomes [21]. This is a model-embedded, clinically-named signal, not a post hoc rationalisation. The 0.4% reduction in AUC is a deliberate and justified trade for this interpretability gain. In medical AI, a model that clinicians can trust and interrogate is more valuable than one that cannot be explained, regardless of a marginal AUC advantage.

C. Why BiLSTM+TII Over TE?

The TE has AUC 0.995 compared to BiLSTM+TII AUC 0.994, a difference of AUC change = 0.0011 that is statistically insignificant in this test set. The TE architecture is much more complex, with two stacked multi-head attention blocks, positional encoding and more parameters. Its attention weights, though visualisable, do not apply universally among patients, and do not relate to named clinical features, limiting their utility as independent explanations of clinical phenomena. The TE also has increased computational requirements during training and inference time. As the BiLSTM+TII is equally able to achieve the same discrimination using a less complex architecture with a high level of interpretability, it is more realistic and principled in regards to actual clinical application.

IX. LIMITATIONS

Future work will include external validation on multi-center datasets to improve generalizability. Single-center generalisation. MIMIC-IV is based on one academic medical center and covers largely an ICU-admitted population. The trained models might not be generalised to outpatient or community-based CKD populations

with varying distributions of disease severity. Before clinical implementation, external validation on independent cohorts like NHANES or regional nephrology registries would be required.

Selection bias. Patients who had less than three visits were eliminated to facilitate sequence modelling and TII calculation. This over-representationally eliminates those with earlier-stage or less closely followed CKD, which may cause learned models to be biased to higher-risk patients. On this basis, about 20-30 percent of the original patient population was screened out. Gradient attribution scope.

Gradient saliency maps offer local explanations, but do not necessarily imply global validity. This was tackled by introducing SHAP to global RF interpretability and population-level averaged saliency to the BiLSTM+TII to offer both local and global coverage. [9].

Transformer comparison scope. Clinical BERT variants were excluded as they require free-text clinical note pretraining and are architecturally incompatible with structured numerical time-series input. The Temporal Fusion Transformer (TFT), while theoretically well-suited to multi-horizon time-series forecasting, was not evaluated in this study as the task is binary progression classification rather than multi-step ahead forecasting — the domain for which TFT's gating and variable selection mechanisms are primarily designed. The added architectural complexity and parameter count of TFT would also make interpretability comparison with BiLSTM+TII less direct. Comparison with TFT on a multi-horizon CKD forecasting formulation remains a concrete direction for future work.

X. FUTURE SCOPE

Several avenues for future work can be identified. First, the current scope of the TII framework is based on three biomarkers. Future work could focus on determining if the accuracy of the model can be further enhanced if the framework is extended to include the urine albumin-to-creatinine ratio, potassium, phosphate, and parathyroid hormone, which are commonly monitored and have independent stage-specific prognostic implications under the current KDIGO framework [16].

Second, the current model architecture can be compared to other sequence models, such as the Temporal Fusion Transformer and other clinical BERT variants pre-trained on substantial amounts of electronic health record data. The self-attention mechanism of the Transformer architecture is theoretically well-suited to handle long-range dependencies and irregularly spaced time series data without the need for padding, which may be beneficial for patients with long visit history.

Third, while the existing explainability module uses gradient-based methods, in the future, it would be valuable to investigate model-agnostic techniques such as SHAP and LIME to enable more globally consistent feature importance measures, which can be validated in structured user studies with domain clinicians.

This article, as with all machine learning models, validation in new, independent cohorts of outpatients with nephrology disorders is critical for clinical translation. The current study exclusively uses MIMIC-IV, which has a population with an ICU bias; validation in community-based cohorts, such as the CKD Biomarker Consortium or National Health and Nutrition Examination Survey, would be valuable to determine whether TII generalizes to different clinical and population groups. Federated learning would allow this

. while ensuring data privacy, which is a significant barrier to clinical translation.

Finally, while this article's development of a dashboard using the Flask platform is intended as a proof of concept for how this article's predictive service could be used in practice, actual engineering work should be undertaken to connect this predictive service to hospital Electronic Health Record platforms using HL7 FHIR APIs, in order to facilitate real-time risk stratification at the point of care without the need for data entry. Longitudinal alert systems could also be used to notify practitioners when a patient's TII exceeds a threshold value for a population.

XI. CONCLUSION

This work bridges the gap between high-performance temporal modeling and clinically interpretable decision support, which remains a key challenge in healthcare AI. The current paper presented a BiLSTM model of CKD progression prediction that is augmented with the Temporal Instability Index, a new, clinically-driven feature that measures the volatility of biomarkers over longitudinal patient data. The framework, which was trained and validated on MIMIC IV v3.1 (40,000 patients, 240,000 visits), was compared to a baseline of a Random Forest, a standard BiLSTM, and a Transformer Encoder in a four-way evaluation on a shared held-out test set.

All three deep models performed significantly better than the RF baseline (meaning, there is a difference in the AUC of about 0.20), which proves that predicting CKD progression requires a temporal modelling. BiLSTM+TII is the most successful deep model with AUC-ROC of 0.994 and F1 score of 0.963, statistically equal to the plain BiLSTM (AUC 0.998) and TE (AUC 0.995), but is the only model that offers interpretable and named risk signals in the form of TII components that can be easily communicated to The highest score of the model in the recall (0.973) is another indication that the model is clinically suitable where the most expensive mistake is the omission of a progression event. A sequence length ablation experiment verifies the requirement of full visit history to perform optimally and independent convergence of SHAP and gradient saliency results confirms the selection of eGFR, creatinine, and SBP as the TII biomarkers.

The fundamental contribution is not just a metric enhancement but a philosophical case of clinical interpretability as a first-class design goal. The TII represents a statistically significant and directly knowledge-based biomarker volatility signal. In clinical decision support applications where predictive accuracy is less important than trust, auditability, and accountability, the BiLSTM+TII is a significant and practical improvement over pure black-box temporal models.

REFERENCES

- [1] Zhu Y, Bi D, Saunders M, Ji Y. Prediction of chronic kidney disease progression using recurrent neural network and electronic health records. *Sci Rep.* 2023 Dec 13;13(1):22091. doi: 10.1038/s41598-023-49271-2. PMID: 38086905; PMCID: PMC10716428.
- [2] L. Men, N. Ilk, X. Tang, and Y. Liu, "Multi-disease prediction using LSTM recurrent neural networks," *Expert Systems with Applications*, vol. 177, p. 114905, 2021.
- [3] Gao, Jing-li & Deng, Hua & Wang, Guo-Dong & Deng, Hui-ling & Feng, Dong-Yi & Wu, Shou-ling & Chen, Shuohua & Zhou, Yan-Feng. (2025). Time-dependent association between progression of arterial stiffness and risk of incident chronic kidney disease: a cohort study in China. *Diabetologia*. 69. 663-673. 10.1007/s00125-025-06609-x.
- [4] Gayathri Hegde M, P Deepa Shenoy, Venugopal KR, Arvind Canchi, A Deep Learning Framework for Chronic Kidney Disease stage classification, *Healthcare Analytics*, Volume 7, 2025, 100398, ISSN 2772-4425, <https://doi.org/10.1016/j.health.2025.100398>.
- [5] Okita J, Nakata T, Uchida H, Kudo A, Fukuda A, Ueno T, Tanigawa M, Sato N, Shibata H. Development and validation of a machine learning model to predict time to renal replacement therapy in patients with chronic kidney disease. *BMC Nephrol.* 2024 Mar 16;25(1):101. doi: 10.1186/s12882-024-03527-9. PMID: 38493099; PMCID: PMC10943785.
- [6] J. Aoki, C. Kaya, O. Khalid, T. Kothari, M. A. Silberman, C. Skordis, J. Hughes, J. Hussong, and M. E. Salama, "CKD progression prediction in a diverse US population: a machine learning model," *Kidney Medicine*, vol. 5, no. 9, p. 100692, Sep. 2023.
- [7] Grams, Morgan E et al. "Predicting timing of clinical outcomes in patients with chronic kidney disease and severely decreased glomerular filtration rate." *Kidney international* vol. 93,6 (2018): 1442-1451. doi:10.1016/j.kint.2018.01.009
- [8] S. Reddy, S. Roy, K. W. Choy, S. Sharma, K. M. Dwyer, C. Manapragada, Z. Miller, J. Cheon, and B. Nakisa, "Predicting chronic kidney disease progression using small pathology datasets and explainable machine learning models," *Computer Methods and Programs in Biomedicine Update*, vol. 6, p. 100160, 2024.
- [9] F. Di Martino and F. Delmastro, "Explainable AI for clinical and remote health applications: a survey on tabular and time series data," *Artificial Intelligence Review*, vol. 56, pp. 5261–5315, 2023.
- [10] T. Anbazhagan and B. Rangaswamy, "Early prediction of CKD from time series data using adaptive PSO optimized echo state networks," *Scientific Reports*, vol. 15, p. 6966, 2025, doi: 10.1038/s41598-025-91028-6.
- [11] F. Khalid *et al.*, "Predicting the progression of chronic kidney disease: A systematic review of artificial intelligence and machine learning approaches," *Cureus*, vol. 16, no. 5, p. e60145, May 2024, doi: 10.7759/cureus.60145.
- [12] Saito H, Yoshimura H, Tanaka K, Kimura H, Watanabe K, Tsubokura M, Ejiri H, Zhao T, Ozaki A, Kazama S, Shimabukuro M, Asahi K, Watanabe T, Kazama JJ. Predicting CKD progression using time-series clustering and light gradient boosting machines. *Sci Rep.* 2024 Jan 19;14(1):1723. doi: 10.1038/s41598-024-52251-9. PMID: 38242985; PMCID: PMC10798962.

- [13] Ayala Solares JR, Diletta Raimondi FE, Zhu Y, Rahimian F, Canoy D, Tran J, Pinho Gomes AC, Payberah AH, Zottoli M, Nazarzadeh M, Conrad N, Rahimi K, Salimi-Khorshidi G. Deep learning for electronic health records: A comparative review of multiple deep neural architectures. *J Biomed Inform.* 2020 Jan;101:103337. doi: 10.1016/j.jbi.2019.103337. PMID: 31916973.
- [14] Johnson AEW, Bulgarelli L, Shen L, Gayles A, Shammout A, Horng S, Pollard TJ, Hao S, Moody B, Gow B, Lehman LH, Celi LA, Mark RG. MIMIC-IV, a freely accessible electronic health record dataset. *Sci Data.* 2023 Jan 3;10(1):1. doi: 10.1038/s41597-022-01899-x. Erratum in: *Sci Data.* 2023 Jan 16;10(1):31. doi: 10.1038/s41597-023-01945-2. Erratum in: *Sci Data.* 2023 Apr 18;10(1):219. doi: 10.1038/s41597-023-02136-9. PMID: 36596836; PMCID: PMC9810617.
- [15] Takkavatakarn, Kullaya et al. "Machine learning models to predict end-stage kidney disease in chronic kidney disease stage 4." *BMC nephrology* vol. 24,1 376. 19 Dec. 2023, doi:10.1186/s12882-023-03424-7
- [16] H. K. Bhachu *et al.*, "Use of the kidney failure risk equation to inform clinical care of patients with chronic kidney disease: A mixed-methods systematic review," *BMJ Open*, vol. 12, p. e055572, 2022, doi: 10.1136/bmjopen-2021-055572.
- [17] Ganie SM, Dutta Pramanik PK, Mallik S, Zhao Z (2023) chronic kidney disease prediction using boosting techniques based on clinical parameters. *PLoS ONE* 18(12): e0295234. <https://doi.org/10.1371/journal.pone.0295234>
- [18] Sendak, Mark & Elish, Madeleine & Gao, Michael & Futoma, Joseph & Ratliff, William & Nichols, Marshall & Bedoya, Armando & Balu, Suresh & O'Brien, Cara. (2019). "The Human Body is a Black Box": Supporting Clinical Decision-Making with Deep Learning. 10.48550/arXiv.1911.08089.
- [19] Wu, TingTing et al. "Logistic regression technique is comparable to complex machine learning algorithms in predicting cognitive impairment related to post intensive care syndrome." *Scientific reports* vol. 13,1 2485. 11 Feb. 2023, doi:10.1038/s41598-023-28421-6
- [20] Khalid, Fizza et al. "Predicting the Progression of Chronic Kidney Disease: A Systematic Review of Artificial Intelligence and Machine Learning Approaches." *Cureus* vol. 16,5 e60145. 12 May. 2024, doi:10.7759/cureus.60145
- [21] Shlipak, Michael G et al. "The case for early identification and intervention of chronic kidney disease: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference." *Kidney international* vol. 99,1 (2021): 34-47. doi:10.1016/j.kint.2020.10.012
- [22] Kim S, Kim W, Park RW. A Comparison of Intensive Care Unit Mortality Prediction Models through the Use of Data Mining Techniques. *Healthc Inform Res.* 2011 Dec;17(4):232-43. doi: 10.4258/hir.2011.17.4.232. Epub 2011 Dec 31. PMID: 22259725; PMCID: PMC3259558.
- [23] W. Lin, A. Chang, D. Crews *et al.*, "Family history of kidney failure, APOL1 risk variants, social determinants of health, and risk of CKD progression: findings from the CRIC study," *American Journal of Kidney Diseases*, vol. 87, pp. 529–539, 2026.
- [24] Sasaki, Takaya et al. "Variability in eGFR and the Risk of Adverse Kidney Outcomes and All-Cause Mortality: A Retrospective Observational Study Based on National General Practice Dataset." *Kidney360* vol. 7,2 (2026): 312-320. doi:10.34067/KID.0000000988
- [25] Q. Abbas, W. Jeong, and S. W. Lee, "Explainable AI in clinical decision support systems: A meta-analysis of methods, applications, and usability challenges," *Healthcare*, vol. 13, no. 17, p. 2154, 2025, doi: 10.3390/healthcare13172154.